

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 02

Lab Report Name : Straight Cable Configuration using CAT6 cable

Course Title : Computer Networks Lab

Course Code : ICT-3202

Submitted By	Submitted To
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Date of Performance :

Date of Submission :

Experiment No: 02

Experiment Name: Straight Cable Configuration Using CAT6 Cable

Objectives:

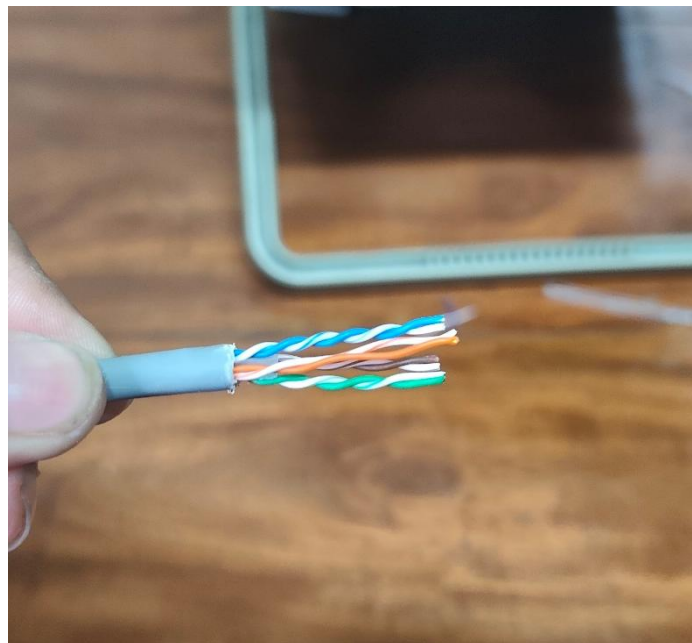
1. To understand the concept and structure of a CAT6 cable.
2. To learn the standard wiring color sequence used for straight-through cable connections.
3. To practically prepare a straight cable using CAT6 cable, RJ45 connectors, and a crimping tool.
4. To test the connectivity of the prepared straight cable using a cable tester.
5. To understand the practical application of straight cables in networking (connecting dissimilar devices, e.g., PC to Switch/Router).

Necessary Materials and Software:

1. CAT6 Cable (UTP)
2. RJ45 Connectors (2 pieces)
3. RJ45 Crimping Tool
4. Cable Cutter/Stripper
5. Cable Tester (LAN Tester)

Procedure:

1. Cut the Cat6 cable to the desired length using a cable cutter.
2. Using the stripper on the crimping tool, remove about 1 inch of the outer jacket from both ends of the cable, exposing the four twisted pairs (8 wires).



3. Untwist the pairs and arrange the eight wires in order according to the T568B standard (same sequence on both ends, since this is a straight cable):

Pin 1: White-Orange

Pin 2: Orange

Pin 3: White-Green

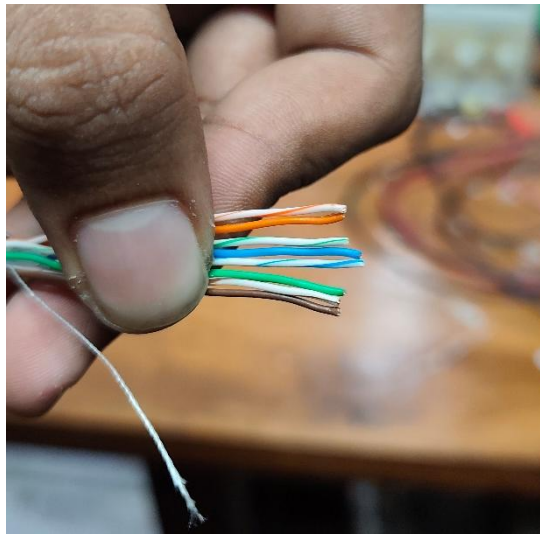
Pin 4: Blue

Pin 5: White-Blue

Pin 6: Green

Pin 7: White-Brown

Pin 8: Brown

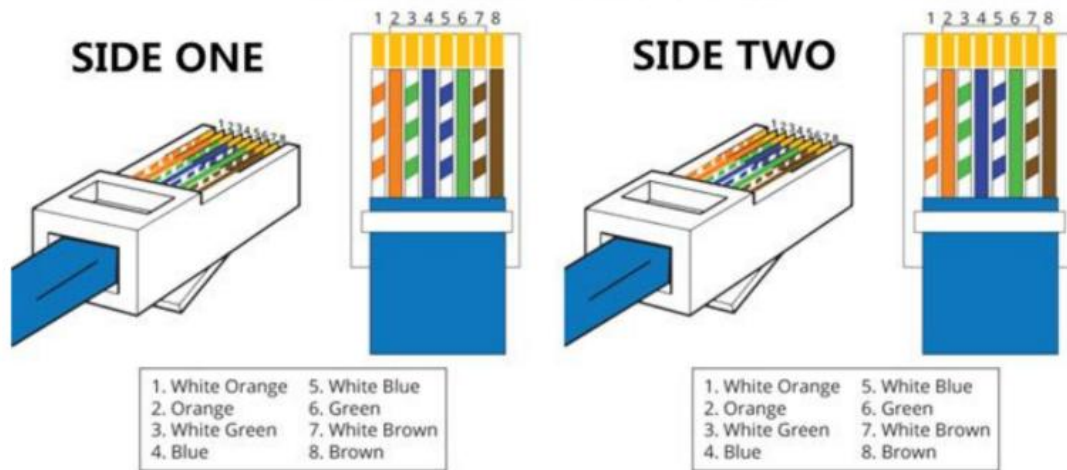


4. Straighten and flatten the wires, then trim them evenly to about 0.5 inch so they fit properly into the RJ45 connector.

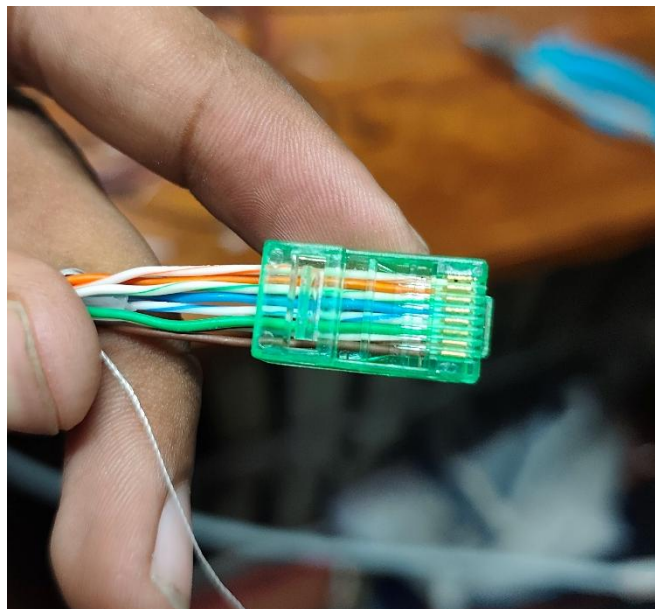


Straight Cable Configuration

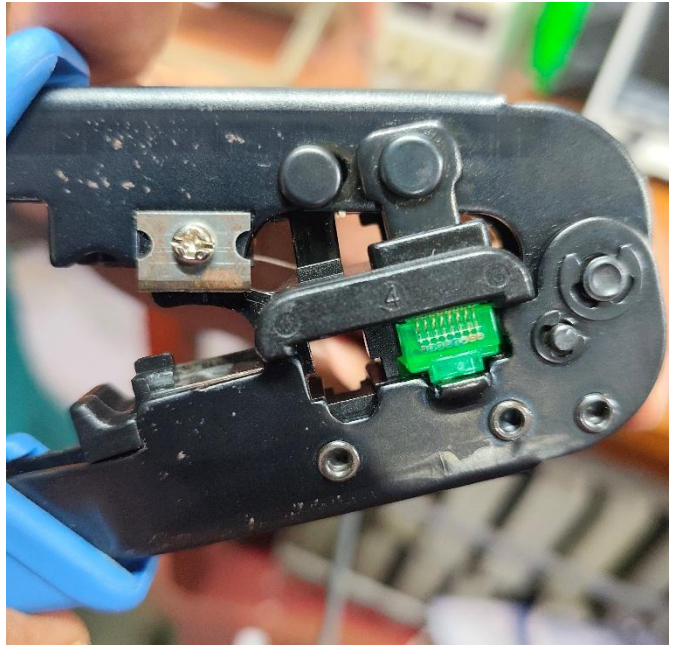
STRAIGHT-THROUGH



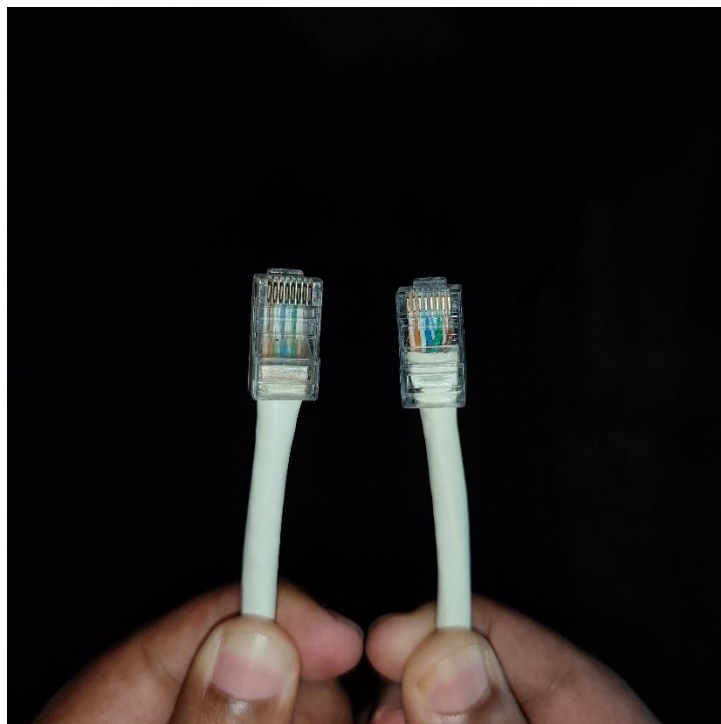
5. Insert the wires into the RJ45 connector in the correct order, making sure each wire reaches the front end of the connector and the outer jacket is inserted slightly into the connector for strain relief.



6. Place the connector into the crimping tool and press firmly to crimp the connector, securing the copper pins onto the wires.



7. Repeat the previous 2-6 steps for the other end of the cable, using the same T568B wiring order (straight-through configuration).



8. Insert both ends of the finished cable into a cable tester to check for correct pin-to-pin connectivity (all 8 LEDs on both ends of the tester should light up in sequence).





Conclusion:

In this experiment, a straight-through Cat6 cable was successfully prepared by wiring both ends of the cable using the same T568B standard color sequence. The cable was tested using a cable tester, confirming correct pin-to-pin connectivity. This type of cable is used to connect dissimilar networking devices, such as a computer to a switch or router. Through this experiment, practical skills in cable termination, crimping, and network cable testing were gained, which are fundamental for setting up wired LAN connections.